COLLEGE OF COMPUTING AND INFORMATICS
PUTRAJAYA CAMPUS

ASSIGNMENTS NO. 2:
DEVELOP YOUR SECURED SYSTEM

SEMESTER 1 2026/2027

PROGRAMME : Bachelor of Computer Science (Cyber Security) (Hons.)
Bachelor of Software Engineering (Hons.)

SUBJECT CODE : CCSB5113

SUBJECT : SOFTWARE SECURITY

START DATE : WEEK 7 (5 Oct – 9 Oct)
DUE DATE : Submission of Report to Brighten: 15 Nov 2026 (1159PM)
: Presentation and Demo: during each presentation day

NAME OF GROUP : MARVELLOUS GROUP

NAME OF MEMBERS :

No.	Full Name	ID Student	Class Group	Task
1.	Khairunnisa bt Osman	12345678	01B	Q1

INSTRUCTIONS TO STUDENTS:

1. Answer **ALL** questions.
2. All answers must be submitted via Brighten.

ASSIGNMENTS NO. 2:
DEVELOP YOUR SECURED SYSTEM

Objective

This assignment requires you to develop a secured system based on the risk assessments on the functionality requirements and security requirements that you have identified and assessed in Assignment 1 (for all sections) and Lab 1–Lab 4 (for Section 1AT and 1BT).

Your goal is to demonstrate the final product after applying the Secured Software Development Life Cycle (SSDLC) during the requirement and the design phase.

Task Description

System Development

- Design and develop a working system or prototype that meets the functionality identified earlier.
- Implement appropriate security mechanisms (minimal security requirements: Multi-Factor Authentication (MFA), hashed password, input validation (syntactical and semantic validation), secured from SQL injection and secured from Buffer Overflow and good exception handling.

Output (Report):

1. Project source code or prototype files (in a compressed ZIP folder)
2. Graphical User Interface (GUI) from the user's view and admin view (if required) (in PDF)
   a) Provide updated design architecture after applying security controls. Use your Microsoft Threat Modelling Tool to design the updated architecture.
   b) Explain the functional requirements that have been implemented
   c) Explain how each implemented security feature mitigates the threats or vulnerabilities identified in earlier tasks by mapping back with your risk assessments.
   d) Besides the GUI, you also can highlight certain code snippets, or configuration details if it is part of your security controls.

Output (Presentation):

1. Begin the presentation with a demonstration of SQL Injection and Buffer Overflow vulnerabilities in your system.
2. After the demonstration, present your secured system and explain how the implemented security mechanisms address or prevent these vulnerabilities.
3. Slides are not required. You may present directly using your system and demonstrate the relevant features and security controls.

Rubric (Total: 100 Marks)

Criteria	Description	Marks
System Functionality	System meets identified functional requirements and performs as intended.	25
Security Implementation	Security features implemented effectively based on identified requirements.	25
Documentation Quality	Clarity, completeness, accuracy and quality of system report	20
Presentation and Submission	Organized, well-presented, and submitted on time.	20

Rubric: System Functionality (25 Marks)

Level	Description	Marks Range
Excellent (21–25 marks)	The system is **fully functional**, meeting **all identified functional requirements** accurately. All main features operate as intended with **no major bugs or errors**. The system demonstrates **smooth integration** between components and includes **additional useful enhancements** beyond the basic requirements.	21–25
Good (16–20 marks)	The system meets **most functional requirements**. Core features work correctly, but **minor issues or incomplete features** may exist. Overall system performance is stable and meets user expectations.	16–20
Satisfactory (11–15 marks)	The system meets **some functional requirements**, but several core functions are **incomplete or partially working**. The system demonstrates basic understanding of the functionality but lacks consistency or completeness.	11–15
Poor (6–10 marks)	The system demonstrates **limited functionality**. Many core features are missing or **do not work as intended**. Frequent errors occur, and user interaction is significantly hindered.	6–10
Very Poor (0–5 marks)	The system is **non-functional** or does not meet any of the stated functional requirements. Little to no evidence of working features.	0–5

Rubric: Security Implementation (25 Marks)

Level	Description	Marks Range
Excellent (21–25 marks)	Security features are **comprehensively implemented** and **align closely with the identified security requirements**. Demonstrates **strong understanding** of security principles such as authentication, authorization, encryption, input validation, and error handling. Implementation **effectively mitigates** the identified threats and vulnerabilities. **No major security flaws** detected.	21–25
Good (16–20 marks)	Most identified security requirements are implemented correctly. Demonstrates **good understanding** of security mechanisms, though **minor gaps or weaknesses** exist. The system shows **reasonable protection** against common threats.	16–20
Satisfactory (11–15 marks)	Some security requirements are implemented, but **coverage is partial or inconsistent**. The system shows **basic awareness** of security needs but may **lack depth** or have **several weaknesses** that could expose vulnerabilities.	11–15
Poor (6–10 marks)	Limited implementation of security features. Security controls are **incomplete or improperly configured**, leaving the system **vulnerable** to threats. Minimal linkage to identified requirements.	6–10
Very Poor (0–5 marks)	**No effective security features** implemented, or implementation is incorrect. Security considerations are **absent or entirely superficial**. System remains **highly insecure**.	0–5

Rubric: Documentation Quality (20 Marks)

Level	Description	Marks Range
Excellent (17–20 marks)	Documentation is **clear, comprehensive, and professionally written**. Includes all required sections: system overview, architecture, functional and security requirements, implementation details, testing results, and reflection. Content is **accurate, well-organized**, and **free from grammatical or formatting errors**. Demonstrates **deep understanding** of both technical and security aspects.	17–20
Good (13–16 marks)	Documentation is **mostly complete and clear**. Covers most key sections with **minor omissions or inconsistencies**. Information is **accurate and understandable**, though may lack polish or depth in certain areas. Few language or formatting issues.	13–16
Satisfactory (9–12 marks)	Documentation is **partially complete** with some missing or underdeveloped sections. Information may be **unclear or repetitive**. Some inaccuracies or organizational issues are	9–12

Level	Description	Marks Range
	present. Shows **basic understanding** but lacks depth or coherence.	
Poor (5–8 marks)	Documentation is **unclear, incomplete, or disorganized**. Key sections are **missing or inaccurate**. Limited explanation of design, implementation, or testing. Writing quality and structure **hinder understanding**.	5–8
Very Poor (0–4 marks)	**No proper documentation** provided, or content is mostly irrelevant, incorrect, or unreadable. Lacks clarity, organization, and completeness.	0–4

Rubric: Presentation (20 Marks)

Level	Description	Marks Range
Excellent (17–20 marks)	Presentation (if applicable) is **clear, confident, and engaging**, effectively demonstrating the system's functionality and security features. Strong teamwork and time management are evident.	17–20
Good (13–16 marks)	Presentation is clear but may lack depth or flow. System demonstration is understandable with slight technical or communication issues.	13–16
Satisfactory (9–12 marks)	Presentation is understandable but lacks confidence, clarity, or structure. Some files or deliverables are missing, or the system demonstration is limited.	9–12
Poor (5–8 marks)	Presentation is unclear, unprepared, or shows limited understanding of the project. Missing key deliverables or not following submission guidelines.	5–8
Very Poor (0–4 marks)	Little to no effort shown in communication or organization.	0–4